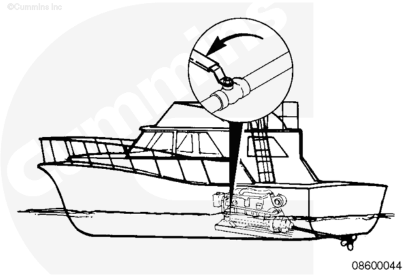
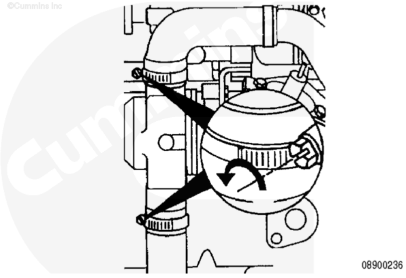
Trainings

Preparatory Steps

Shut off the sea water inlet valve on the vessel hull.



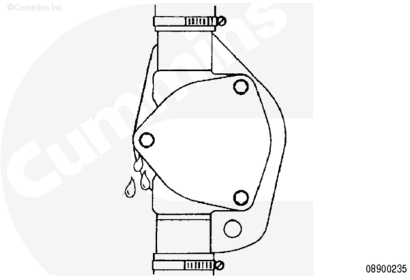
Remove the inlet and outlet hose and drain the water from the pump.



Remove

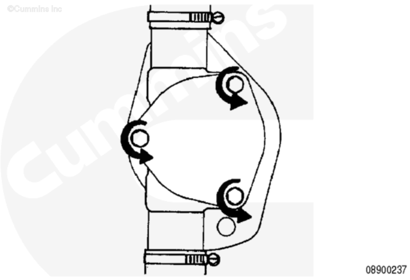
⚠ CAUTION ⚠

If the impeller has failed and pieces are missing, all pieces must be retrieved. The engine heat exchanger, gear oil cooler, and sea water aftercooler (if equipped) must be flushed. Refer to the procedures for flushing these components in other headings of this section. Failure to do so can result in overheating and damage to engine can occur.



Impeller debris can also drop into the inlet piping. Make sure all debris is removed before installing a new impeller; otherwise, additional impeller failures or engine overheating will occur.

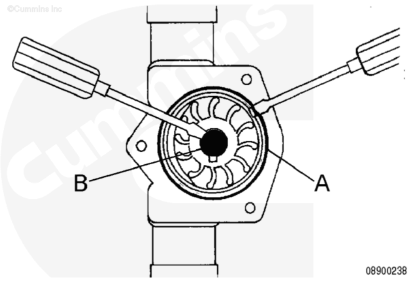
Remove the capscrews. Lift off the cover.



Use a small screwdriver to remove the o-ring (A).

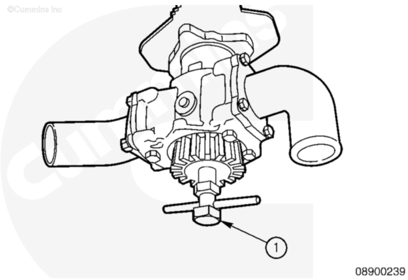
Use a small screwdriver to remove the inner cap (B).

Clean the o-ring groove.



⚠ CAUTION ⚠

Do not pry against the pump housing to remove the impeller as this can cause damage to the liner.



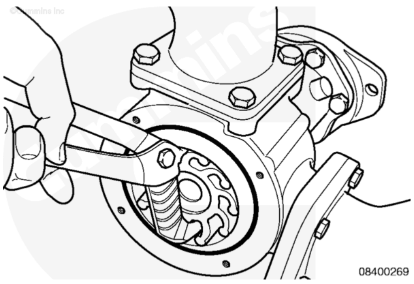
Be sure to note the direction of impeller fins for proper re-installation. Mark the outer surface.

An impeller removal tool is available from Sherwood Pumps, Part Number 23631.

If the impeller is equipped with a threaded insert, use the special tool or a ¾-NFT bolt (1) to insert in the impeller to pull the impeller out.

Be sure to note the direction of impeller fins for proper re-installation. Mark the outer surface.

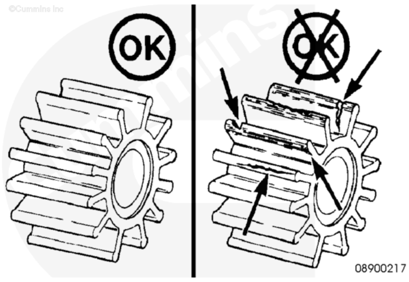
If the impeller does **not** have a threaded bore, grasp the hub of the impeller with pliers and remove the impeller from the impeller bore.



Clean and Inspect for Reuse

Inspect for damage such as rips, tears, chunks of material missing, or wear on the edges of the blades.

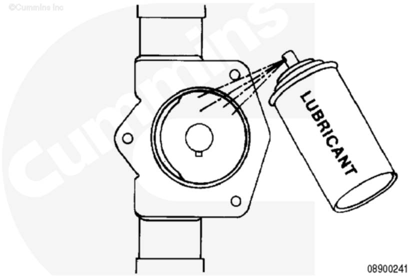
Replace as necessary.



Clean the internal pump surfaces.

Lubricate the housing with silicone or glycerine non-petroleum-based lubricant. Petroleum-based lubricant will damage the rubber impeller.

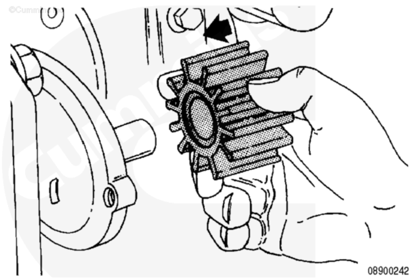
If non-petroleum-based lubricant is **not** readily available, use soapy water to ease installation.



Install

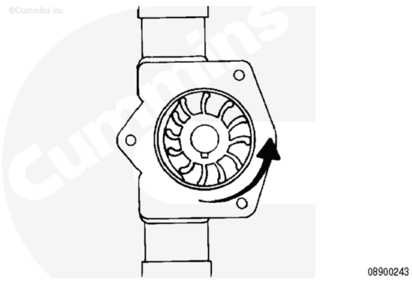
If the impeller is in good shape and will be reused, install it in the same direction from which it was removed. Refer to the mark you made during removal.

If the impeller was **not** marked and the original rotation or direction can **not** be determined, replace the impeller with a new one.

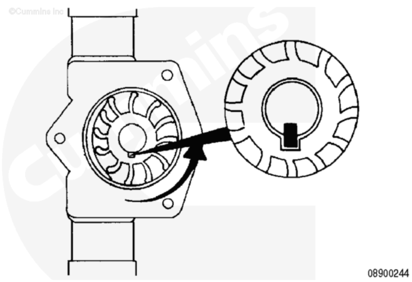


An oil filter strap wrench or even plastic wire straps can be used as an installation aid to hold the vanes.

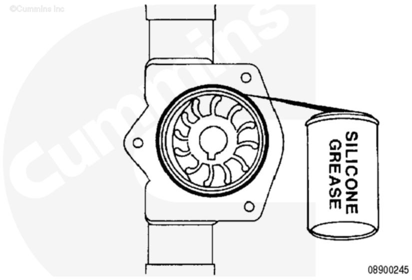
Guide the impeller into the housing, twisting it **counterclockwise** as it is advanced so that the vanes will be deflected in the proper direction.



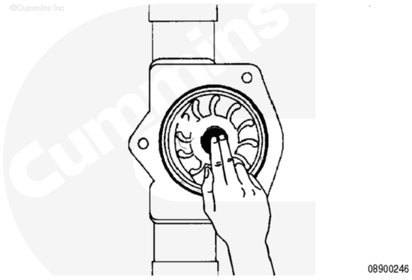
Continue to turn the impeller while pushing it into the housing. It will slide all the way in when the keyway lines up with the key.



Insert the new o-ring into the impeller housing. Use a little silicone grease to hold it in place.



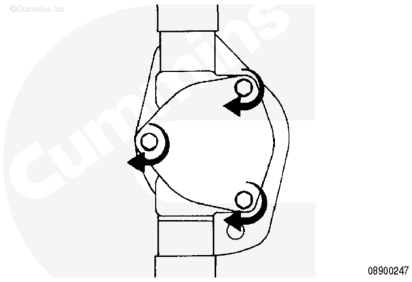
Install the rubber impeller cap into the center hub of the impeller.



Install a new gasket, cover plate, and capscrews.

Tighten the capscrews.

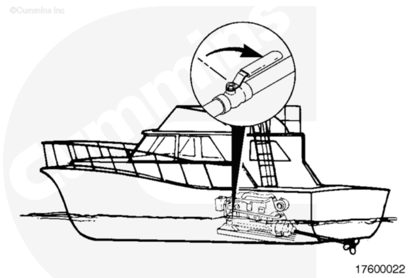
Torque Value: 24 n•m [212 in-lb]



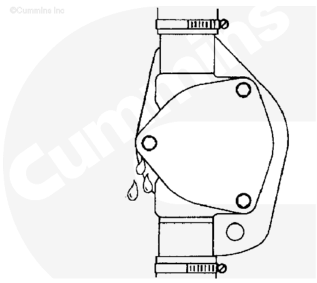
Finishing Steps

Install the inlet and outlet hose to the pump.

Open the sea water inlet valve and check for leaks.



If you have a wet exhaust system, start the engine and check for water flow from the exhaust. Also, check the sea water pump and plumbing for leaks.



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